

Matrices - Summary

* Dimensions

- * A matrix with i rows & j columns is written as a $i \times j$ matrix
- called a i by j matrix -

* Matrix Addition

- * two matrices can only be added when they have exactly the same dimensions. (same no. of rows & columns)
- * matrix addition is performed entry by entry, for matching positions.

$$\text{if } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ and } B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

$$\text{then } A+B = \begin{bmatrix} a+e & b+f \\ c+g & h+d \end{bmatrix}$$

- * this follows basic rules $\Rightarrow A+B = B+A$

$$(A+B)+C = A+(B+C)$$

$$A+O = A$$



this, if 2×2 would be like $= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

* Matrix Subtraction

- Very identical to addition, but you subtract instead of adding.

$$A - B = \begin{bmatrix} a-e & b-f \\ c-g & d-h \end{bmatrix}$$

- common things to keep in mind $\Rightarrow A - B = A + (-B)$

$$A - B \neq B - A$$

$$B - A = -(A - B)$$

* Scalar multiplication

$$\text{if } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ then } kA = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

- for expressions like $3A - 5B$

1) find $3A$

2) find $5B$

3) perform matrix subtraction

* Matrix Multiplication

- this should be performed using rows from first matrix & columns from the second matrix.

$$(AB)_{ij} = \text{row } i \text{ of } A \times \text{column } j \text{ of } B$$

- basically multiply corresponding values & add their results.

$$A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}, \quad B = \begin{bmatrix} g & h \\ i & j \\ k & l \end{bmatrix}, \quad \text{lets find } AB$$

$$AB \Rightarrow \begin{bmatrix} g & h \\ i & j \\ k & l \end{bmatrix} \quad \text{① write them down like this}$$

$$\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} m & n \\ p & q \end{bmatrix}$$

$$m = ag + bi + ck \quad \leftarrow \text{② calculate}$$

$$n = ah + bj + cl \quad m, n, p, q$$

$$p = dg + ei + fk$$

$$q = dh + ej + fl$$

$$\therefore AB = \begin{bmatrix} m & n \\ p & q \end{bmatrix} \quad \leftarrow \text{③ this is the final result.}$$

* Modular Arithmetic

- when reducing a matrix modulo n , reduce every entry separately.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \pmod{n} = \begin{pmatrix} a \pmod{n} & b \pmod{n} \\ c \pmod{n} & d \pmod{n} \end{pmatrix}$$

* Determinant of a 2×2 matrix

$$\text{if } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ then, } \det(A) = ad - bc.$$

* Cofactors

* use this formula $\Rightarrow C_{ij} = (-1)^{i+j} \det(A_{ij})$.

* Generalized Example $\Rightarrow$

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \Rightarrow C_{22} = (-1)^{2+2} \det \left(\begin{bmatrix} a & c \\ g & i \end{bmatrix} \right)$$

← ignore the stuff you selected here.

* to get C of A , do this repeatedly to each and every element and write:

$$C \text{ of } A = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

* Determinant of a 3×3 matrix

* get any row that you think would be the easiest for you to work with, and point this $\Rightarrow$

$$\det(A) = gC_{31} + hC_{32} + iC_{33}$$

↖ You need to have the cofactors.

* Adjugate

$$\text{adj } A = [C(A)]^T \quad \leftarrow \text{this is called the transpose of a matrix}$$

$$C(A) = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

② swap the corresponding parts accordingly

① use this line as an axis

$$\therefore [C(A)]^T = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} = \text{adj}(A).$$

* Matrix Inverse

$$A^{-1} = \frac{1}{\det(A)} \times \text{adj}(A) \quad \leftarrow \text{formula.}$$

where A is a matrix, A^{-1} is its inverse and I is the identity matrix,

$$AA^{-1} = A^{-1}A = I$$

* Matrix Inverses Modulo n

for a matrix to be invertible modulo n , its determinant must have a multiplicative inverse modulo n ,

when $\gcd(\det A, n) = 1$,

$$\text{then, } A^{-1} \equiv (\det A)^{-1} \text{adj}(A) \pmod{n}$$

- and every operation is performed in a modular arithmetic way.
- There is no division in modular arithmetic. Instead, you multiply by the modular multiplicative inverse.

* Reflection

The most important thing I learnt from this module was how to perform different kinds of mathematical operations on matrices, calculate determinants, cofactors, find inverses and use modular arithmetic with matrices.

I have already studied most of this in high school mathematics. Matrix addition, subtraction and scalar multiplication uses regular arithmetic. Matrix multiplication uses multiplication and addition of values calculated using rows and columns. Determinants and cofactors also use basic algebra, multiplication, subtraction and negative numbers. Modular arithmetic as the name suggests is related to finding remainders.

I think the course team wants us to be familiar with matrices because it might help students in understanding cryptographical concepts at a foundation level. A lot of cryptographical concepts deal with discrete mathematics and matrices play a big role in it too.





